Mahmoud Alyosify

Reinforcement Learning Engineer

New Cairo, Egypt · Open to relocation and remote · +20 114 555 7959 · mahmoud.alyosify@gmail.com
linkedin.com/in/mahmoudalyosify · github.com/MahmoudAlyosify · mahmoudalyosifysite.github.io

SUMMARY

Reinforcement learning engineer who builds agents for **decisions with a real cost**. Built a cloud-autoscaling environment and trained PPO and DQN agents to balance cost and latency against the policies teams actually run. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), with graduate coursework in reinforcement learning.

SKILLS

Reinforcement learning: PPO, DQN, custom Gymnasium environments, reward design, simulation
Machine learning: PyTorch, TensorFlow, NumPy, Pandas
Agents: LangGraph, multi-agent systems
Engineering: Python, Docker, AWS, Git, Linux

PROJECTS

RL-Cloud-Autoscaler — deep RL for cloud scaling (Python, Gymnasium, PPO, DQN) 2026

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.
- Designed the environment from scratch: state, actions and a reward that penalises both overspending and slow responses.
- Compared both agents with static allocation and threshold-based autoscaling, the policies real teams run.

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) 2026

- Treats generation as a stop-or-continue decision: a stop rule with a completeness gate makes three open LLMs generate **47.6% fewer tokens**, with no answer cut off.

HORUS Sentinel — multi-agent intelligence platform (Python, LLM agents, knowledge graph)

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo Oct 2026 – Present

- Built a **LangGraph** agent that routes each question, grades the evidence it finds, rewrites weak queries and checks its own answer.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.

Online Instructor — Udemy & YouTube (Arabic courses), Remote 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · 1st Place, Smart Cities Hackathon 2022 (77 university teams) · Semi-Finalist, Microsoft Imagine Cup 2023

LANGUAGES

Arabic (native) · English (professional working proficiency)